

DEBANJAN RAY

MSc Advanced Computer Science | AI & Software Engineering

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EDUCATION

University of Manchester — MSc Advanced Computer Science

2026 – 2027

Newcastle University — BSc Computer Science, First Class Honours

2023 – 2026

Relevant modules: Computer Vision and AI (84%), Computer Systems Design and Architectures (85%), Foundations of Data Science (86%), Programming Portfolio 1 & 2 (82%, 92%), System and Network Security (77%)

Dissertation — LLM-driven research idea generation pipeline

- Designed a pipeline pairing author publication histories from arXiv to test whether cross-domain context improves the novelty of generated research ideas.
- Evaluated outputs across five dimensions using Gemini 2.5 Flash; applied Mann-Whitney U and Spearman correlation tests to assess a domain-distance gradient hypothesis.

TECHNICAL PROJECTS

Agent Flight Recorder — time-travel debugger for LLM agents

- Built an observability and replay tool for multi-step LLM agent executions, enabling step-by-step state inspection and debugging.
- Benchmarked failure modes and documented them transparently in a written Limitations report, including a fix for confidently-wrong responses on out-of-scope queries via explicit replay-mode labelling and constrained scenario inputs.
- Shipped with README, technical report, and screencast demo.

Synthetic Usability Lab — parallel AI-persona website critique engine

- Built a system where multiple AI personas browse and critique websites in parallel, producing structured usability audits.
- Repurposed as an outreach engine: generates unsolicited, evidence-backed audits of live company websites as cold-outreach pitch artefacts.

PortfolioWebsite—personal engineering site

- Built with Next.js 14, TypeScript, Tailwind CSS and Framer Motion; deployed on Vercel with GitHub auto-deploy and branch preview environments.

ADDITIONAL ACADEMIC PROJECTS

- DevOps web application** — full-stack Flask + PostgreSQL app with CI/CD pipelines, Docker containerisation, and centralised logging via the ELK stack.
- Reliable File Transfer protocol** — guaranteed reliable file delivery over UDP using Java network socket programming.
- PCA on MNIST** — dimensionality reduction and visualisation of handwritten digit features.
- Quiz platform (Java)** — led delivery within a 5-person team; 700+ lines of code, 94.25% mark.
- Relational database** — bus route management system built in MySQL via DBeaver, 94% mark.
- Secure Flask blog** — multi-factor authentication, rate limiting, and access control.

TECHNICAL SKILLS

Languages	Python, C++, C#, Java, Go, C, GDScript
AI / ML	Google Gemini API integration, Google Colab, pandas, Matplotlib
Web & Frameworks	Flask, HTML/CSS, JUnit
DevOps	Docker, Git/GitHub, CI/CD, ELK Stack
Databases	SQL, MySQL, PostgreSQL, MongoDB, DBeaver
Other	Linux/Bash, Microsoft Office Suite

ACHIEVEMENTS

- Verizon Cloud Platform Job Simulation** (Forage), Nov 2024 — designed a hypothetical VPN product evaluated against cloud-native principles (redundancy, resiliency, least-privilege); presented security research findings.
- 1st place**, 'In Search of Young Ruskin Bond' writing competition — short story hand-selected by author Ruskin Bond from 4,000+ entries across 84 schools.

INTERESTS

Executive committee member, Newcastle GameDev Society. Active chess player — represented Newcastle University in an inter-university tournament against Durham.